COP 6819 · ADVANCED INTERNET SYSTEMS

CloudAnalyst Results

How load-balancing policy and data-center provisioning shape response time and cost

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6 simulations · up to 7 user bases · 6 data centers

THE SETUP

One base topology for runs 01–05

6
user bases, UB1–UB6

4
data centers, DC1–DC4

1,000
peak users per user base

60
requests per user per hour

Runs 01–05 each change one thing: the service-broker policy, the VM count per data center, or the request size (100 bytes at baseline). Run 06 tests a load-based policy on a larger map of 7 user bases and 6 data centers.

SCENARIOS

Six runs, one question each

#	Scenario	VMs per DC	What it tests
01	Round Robin (baseline)	15 / 5 / 5 / 30	VMs take requests in turn
02	Throttled	1 / 1 / 1 / 30	Caps concurrent requests per VM
03	Under-provisioned DC	1 / 1 / 1 / 30	Same layout, read for the tail
04	Network latency edge case	20 / 1 / 20 / 20	Only DC2 is weak
05	Cost-sensitive	15 / 15 / 20 / 15	UB3, UB5 requests 100× larger
06	Equally Spread (load-based)	5 / 5 / 5 / 5 / 5 / 4	Least-busy VM; 7 UBs, 6 DCs

UB = user base · DC = data center · VM counts read DC1 / DC2 / DC3 … in order

RESPONSE TIME

Average response barely moves



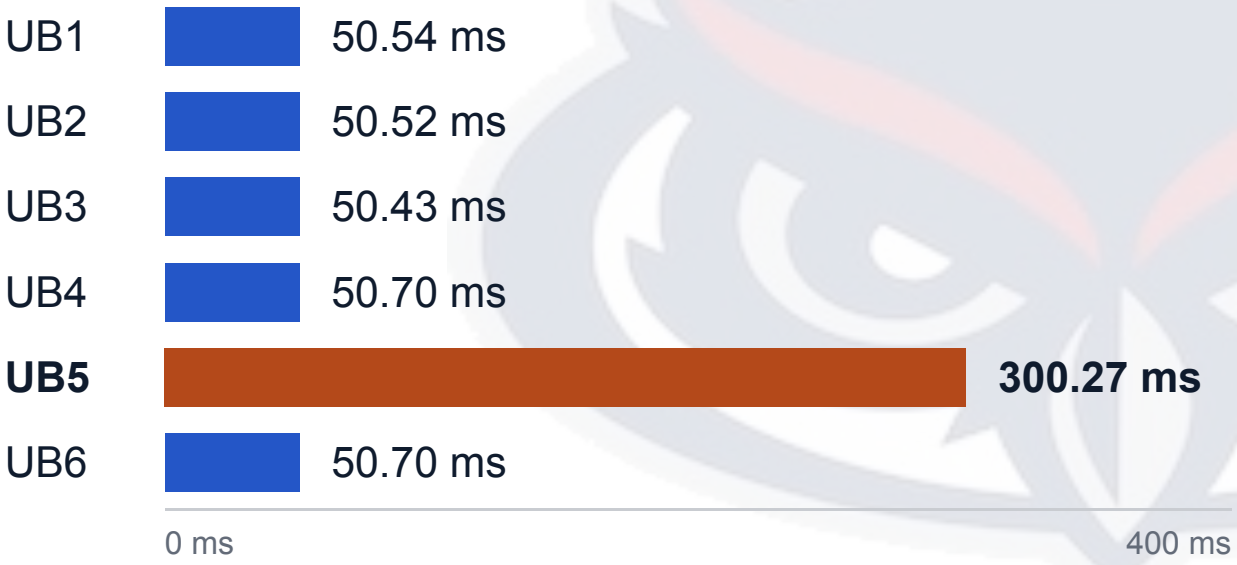
2.85 ms
gap between the fastest and slowest average in runs 01–05, about 3%

3.68 ms
highest average data-center processing time in any run

RESPONSE TIME

Network distance sets the floor

Average response time by user base, baseline run



0.84 ms

average data-center processing time in the baseline, under 1% of the 92.59 ms total

~300 ms

UB5's average in runs 01–05: a fixed geographic floor that no policy changes

LOAD-BALANCING POLICY

Policy trades speed for cost

Round Robin

Fastest average

15 / 5 / 5 / 30 VMs

Average response	**92.59 ms**
DC processing, avg	**0.84 ms**
DC processing, max	**3.40 ms**
Grand total cost	**$161.36**

Throttled

Lowest cost

1 / 1 / 1 / 30 VMs

Average response	**95.43 ms**
DC processing, avg	**3.68 ms**
DC processing, max	**23.76 ms**
Grand total cost	**$108.56**

Throttled saves **$52.80** for **2.84 ms** more average response: pick by SLA or by budget.

RUN 03 · UNDER-PROVISIONED DC

Averages hide the tail

AVERAGE RESPONSE

95.44 ms

Same as the Throttled run (95.43 ms)

MAXIMUM RESPONSE

552.75 ms

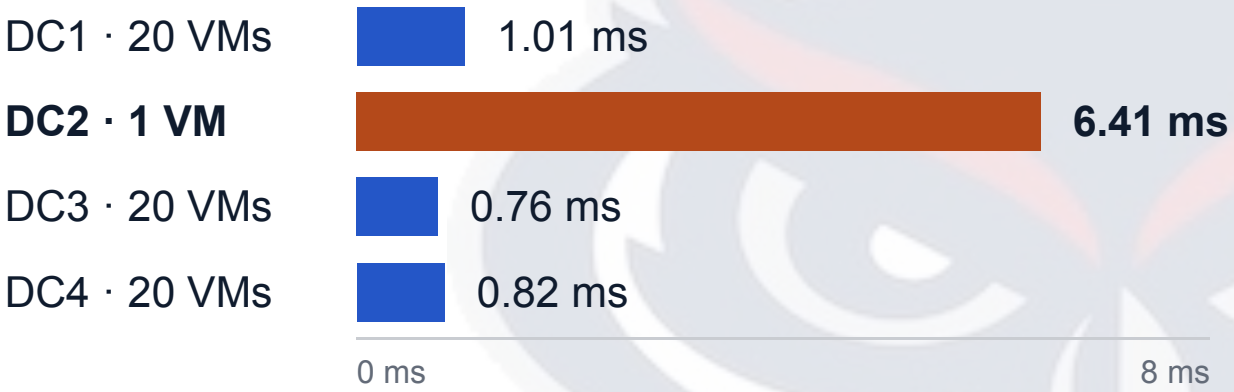
About 38% above the Throttled run's 400.88 ms

Under-provisioning barely moves the mean. The damage lands in the tail, which is exactly what SLA-sensitive workloads feel.

RUN 04 · NETWORK LATENCY EDGE CASE

One weak DC, one slow region

Average data-center processing time, run 04



56.11 ms

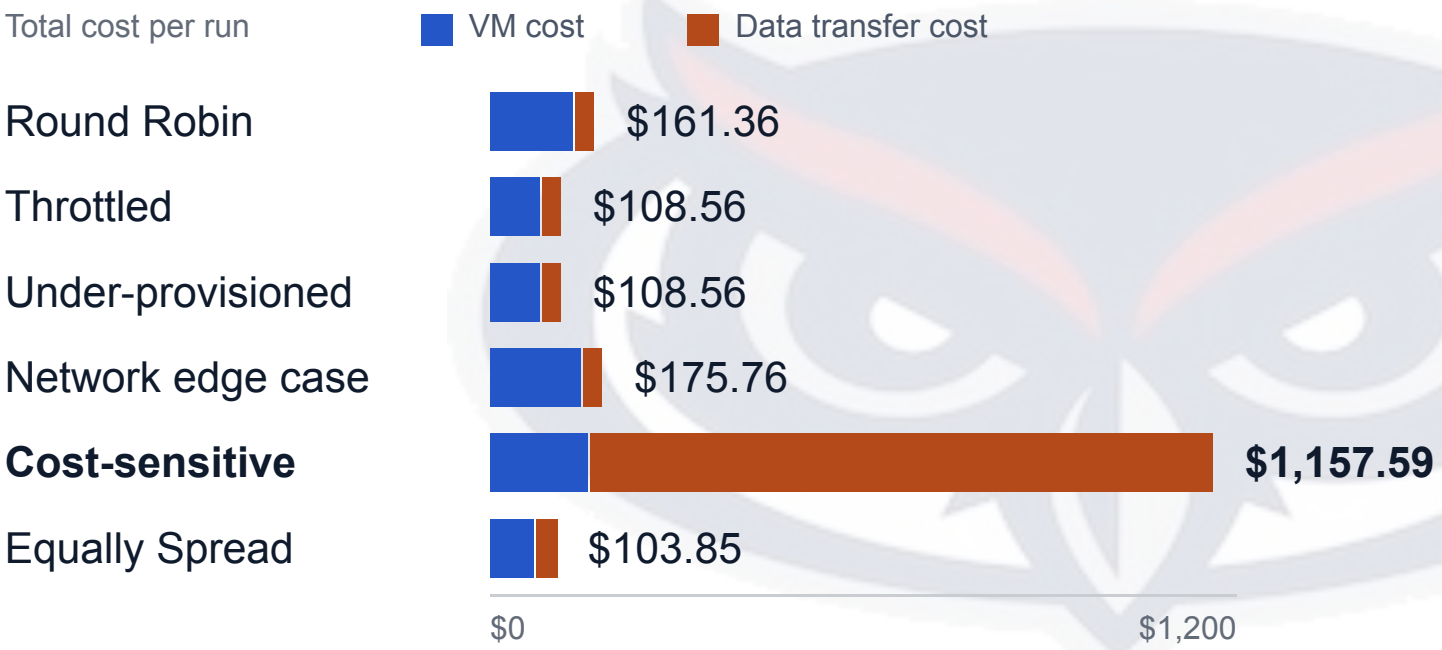
UB2's average response, served by DC2, vs about 50.5 ms for UB1, UB3, UB4 and UB6

Provisioning, not policy

Load balancing cannot make up for a region that lacks capacity.

COST

Data transfer drives the bill



$1,157.59

cost-sensitive grand total, about 7× the $161.36 baseline

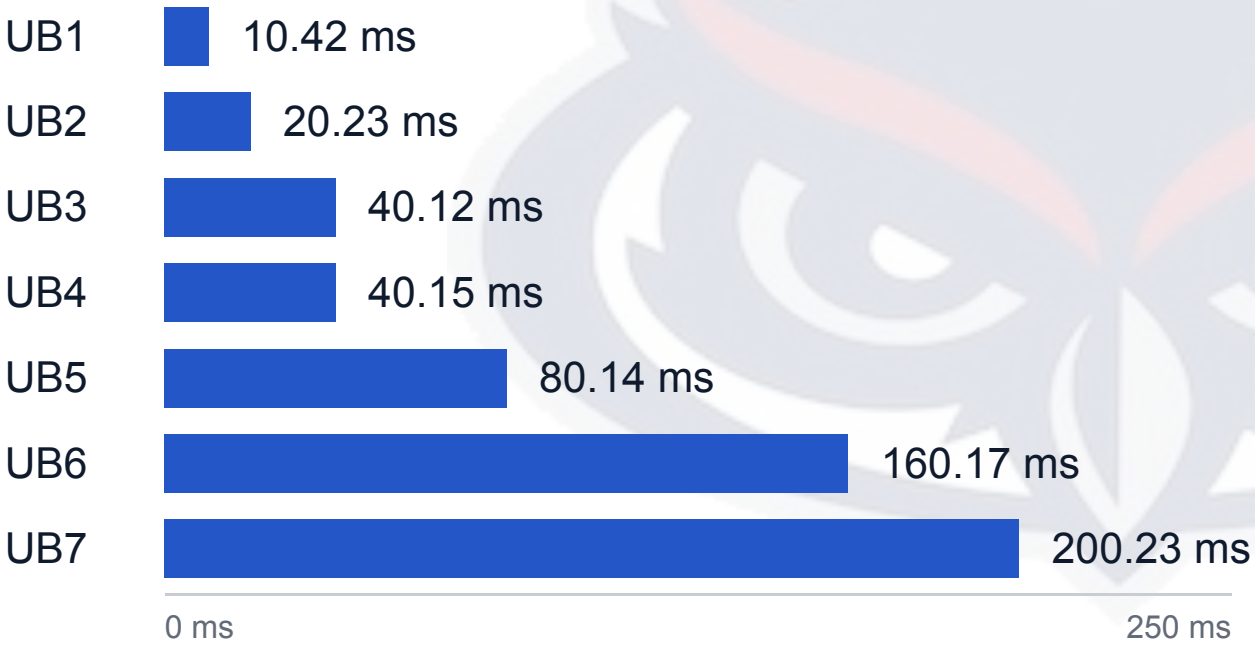
$1,001.58

of that is data transfer, and $981.99 of it lands at DC3

RUN 06 · EQUALLY SPREAD CURRENT EXECUTION LOAD

Distance sets each region's response

Average response time by user base, run 06



0.55 ms

average data-center processing, the lowest of all six runs

78.81 ms

overall average, lower than runs 01–05 because this map has shorter latencies

RUN 06 · LOAD AND COST

Load spreads evenly, but DC6 sits idle



$9.60

DC6's VM bill for a day with no requests: the load balancer only picks VMs inside a data center

Cheapest run: $103.85

29 VMs served 7 user bases, on a different map from runs 01–05.

SUMMARY

All six runs side by side

Scenario	Avg response (ms)	Max response (ms)	Processing avg (ms)	Total cost
Round Robin	92.59	400.88	0.84	$161.36
Throttled	95.43	400.88	3.68	$108.56
Under-provisioned DC	95.44	552.75	3.68	$108.56
Network latency edge case	93.51	401.07	1.76	$175.76
Cost-sensitive	95.20	550.67	0.84	$1,157.59
Equally Spread (run 06) †	78.81	258.31	0.55	$103.85

Blue marks the best value and orange the worst across runs 01–05: fastest average (Round Robin), lowest cost (the 1 / 1 / 1 / 30 layout, runs 02 and 03), worst tail (Under-provisioned DC) and highest cost (Cost-sensitive). † Run 06 uses a larger map of 7 user bases and 6 data centers with shorter latencies, so it is shown in gray and not ranked.

CONCLUSIONS

What the study shows

01 **Network latency dominates the average**

Runs 01–05 all land at 92–95 ms; processing adds only a few milliseconds.

02 **Provisioning drives the tail, not the mean**

Starved DCs push the max to 552.75 ms; the average stays near 95 ms.

03 **Choose the policy by the binding constraint**

On the shared map, Round Robin is fastest at 92.59 ms and Throttled cheapest at $108.56.

04 **Payload size is the biggest cost lever**

Data transfer was $1,001.58 of the $1,157.59 cost-sensitive bill.

05 **Load balancing works inside a data center, not across them**

Equally Spread held processing at 0.55 ms, but idle DC6 still cost $9.60.

THE HABIT TO KEEP

Report the tail, not just the average.

The average masked the worst failure mode in this study. Maximum and 95th-percentile latency belong in every result.

Thank you · Questions?